

Standards-Based Grading in Large-Enrollment Calculus: Practices and Lessons for Students/Faculty

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Outline



- 1 Why Standards-Based Grading (SBG)?
- 2 The Four Pillars
- 3 Lessons Learned

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Why
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(SBG)?

The Four
Pillars

Lessons
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What is SBG?



- Create a list of standards that students should achieve by the end of the course.
- Students' grades are largely based on how many standards they achieve by the end of the term.

Four Pillars

- 1 Clearly-defined standards
- 2 Helpful feedback
- 3 Marks indicate progress
- 4 Reattempts without penalty

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Calculus Courses at Pitt



MATH 0120, 0220, 0230 are dense, fast-paced service courses:

- Thousands of students per year
- 75-person lectures, 25-person recitations taught by TAs
- Prerequisite course(s) for many programs and majors.

Primary objective: teach and evaluate learner competencies, so students can succeed in future courses.

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What I Do



- ~ 10 Big Ideas, 3 attempts per Big Idea.
- Each question is scored either 0 or 1; $3/4$ questions required to pass.
 - First attempt soon after learning the material, graded quickly
 - Second attempt one week later
 - Final attempt at the end of the term
- Most other assignments in the course are structured around the Big Ideas.
- Department-wide final is the last chance to show improvement on the material.

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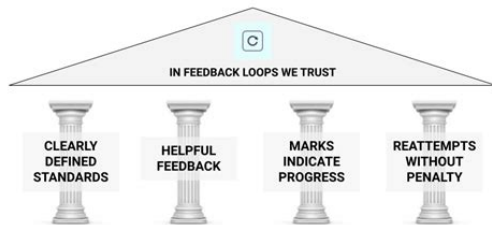
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Clearly-Defined Standards



- By the end of the course, SWBAT...
- Succinct description of discrete skills
- Graded assignments should be clearly aligned with your standards.
- Let your students know about these standards everywhere, all the time!



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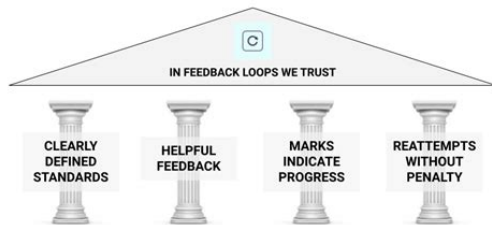
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Helpful Feedback



- For students to improve, they need feedback.
- Comments justifying why points were taken off?
- Having clearly-defined standards makes giving feedback easier.
- Counterfactual explanations also leave a paper trail.



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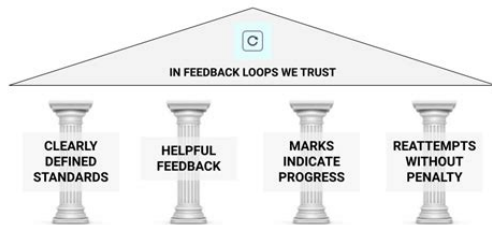
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Marks Indicate Progress



- What does a 7/10 score on a question mean?
- When scoring is 0/1, there are fewer fights over points.
- Separate marks for “perfect” vs. “essentially correct”; both earn 1.
- What is **really** important to this learning outcome?



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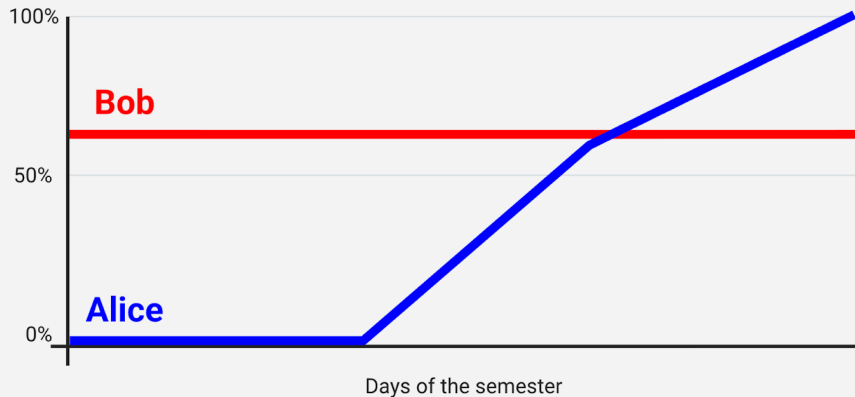
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Student understanding over time



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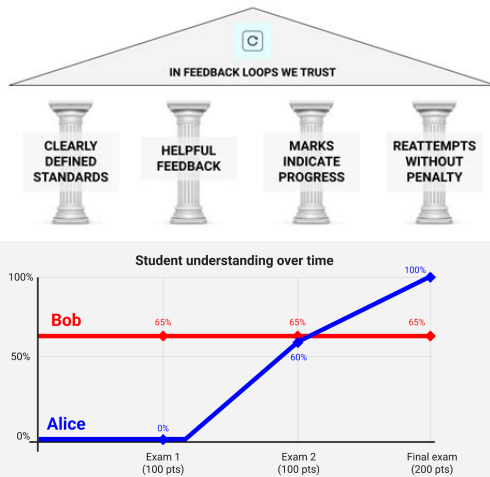
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Why Standards-Based Grading (SBG)?

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- If a student understands all the material by the end of the course, what grade should they earn?
- Example: A “super-final” might replace previous exam scores.



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A Few-Big-Exams Structure Doesn't Hold Up!



- Few big exams means there is less data and less opportunity for growth.
 - What is a student is out sick?
 - What if they are dealing with something very distracting?
 - What if they prioritize a different exam on the same day?
- Partial credit is bad for growth **and** measurement of skills.
- Students do not engage with feedback loops on a regular basis.

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(Pre-Final) Course Grade

To earn a given letter grade, you must satisfy **all** of the requirements in the given row.

Final Grade	Big Ideas Achieved (9+1)	CORE Big Ideas (3)	Engagement Points Earned
A	9	3	200
B	7	3	150
C	6	2	100
D	3	0	0
F	Did not meet all of the requirements for a D		

For information on how the Final Exam can modify your grade, see page 5 of the syllabus.

Keep things simple and reinforce throughout the semester!

Engaging with Struggling Students



- ...who are lacking algebra skills
- ...who are lacking study skills
- ...who are lacking resources
- ...who have test anxiety
- ...who claim to not be math people

Clearly-defined standards can make a world of difference!

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Grading



- Quick turnarounds required, so students have feedback before the next attempt.
- Gradescope helps organize and speed things up.
- No midterm exams means no huge pile of grading all at once.
- With 150 students, it took me 6-8 hours per week to grade attempts (one class' first attempts, one class' second attempts).
- With support, TAs are capable of understanding and using this system.
- More flexibility on attempts makes a zero-make-up policy reasonable.

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Other benefits



- “Good enough” is a high bar, and students will meet that bar!
- This structure respects my students’ diversity
 - Mathematical and non-mathematical skills
 - Majors and interests
 - Tough schedules
- Resistant to generative AI cheating, even with a lot of out-of-class work!

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Advice for Students (with SBG or otherwise)



Align your learning approach with the four pillars!

- What do you need to demonstrate you know?
- If you aren't getting helpful feedback on graded assignments, find another way.
- Do not take partial credit at face value.

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Thank You!



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